Durham University

Examination Paper

Examination Session:	Year:	Exam Code:
May/June	2022	ENGI3341-WE01

Title: (BEng/MEng)

Environmental Engineering 3

Time Allowed:	2 hours
Additional Material provided:	Yes, formula sheet included as part of exam paper
Reporting problems during the exam:	Queries about the instructions for this exam (rubric) should be directed to: eng.boe@durham.ac.uk. If you experience any difficulties uploading your exam answers to Gradescope report these immediately to: eng.boe@durham.ac.uk.

Instructions to Candidates:	Answer ALL questions. Please answer each question on a new page. All workings must be shown. This is an online open-book exam, submission is time-limited to 4 hours within a 24 hrs window. • *The examination attempt must be all your own work. You must not interact with another person (either one-to-one, by telephone, on the internet, in chat rooms, on social media, via smart watches, etc.), or use cheat sites, etc. whilst completing the exam. You must not discuss any examination papers with friends, relatives, current or past students of this or any other University before the end of the relevant 24-hour window.* • *Candidates are expected to answer all questions in their own words. Should close paraphrasing, direct quotation be necessary candidates must acknowledge the source of any material used in their answers (this includes text, images, diagrams, charts, tables and/or graphs). Full referencing is not expected, but a brief indication of the source of the material (e.g. name of author) must be given.*

Revision:	

Q1

A new housing development is to be built on the site of a former electroplating works. A site investigation has identified that the ground strata consists of 1 m of man-made granular fill, overlying 2 m of stiff clay, overlying a bedrock of sandstone. The groundwater level is 2 m below the ground surface and the sandstone bedrock is a known aquifer. The man-made granular fill and the upper 0.25 m of the clay layer have tested positive for the presence of cyanide, nickel, mercury and lead.

(a) What are the main potential pathways and receptors for the contamination within the site.

[40%]

(b) In your own words, describe the main groups of remediation options that might be considered to remediate the site in order to make it suitable and safe for development.

[20%]

(c) The developer would like to treat the contaminated land in-situ in order to avoid expensive removal and disposal costs. What options are available to achieve this and which would you recommend. Justify your suggestion.

[40%]

continued

Q2

Two reservoirs are to be connected by a cylindrical pipe of 2 km length with a slope of 1/2000. The Manning coefficient of the pipe material is 0.02 $m^{-1/3}$ s.

(a) Determine the minimum diameter of pipe (to the nearest 10 cm) required to deliver water at a rate of at least 0.42 m^3 s^{-1}, when the water level difference between the two reservoirs is greater than 3 m.

[40%]

(b) Determine the water depth (to the nearest 10 cm) in such a pipe when the flow rate is 0.10 m^3 s^{-1}, stating any assumptions where appropriate.

[60%]

Q3

A hydrocarbon storage tank leaks into a reservoir with a hydraulic connection to a surrounding aquifer. Elevated benzene concentrations are detected in an observation well located 45 m from the reservoir, 280 days after the leak is believed to have commenced.

The water level in the observation well is 0.92 m below that in the reservoir. The hydraulic conductivity and bulk density of the aquifer material are 1.8 m day^{-1} and 2650 kg m^{-3}, respectively. The K_d partition coefficient for benzene in the aquifer material is 0.038 cm ^{3} g^{-1}. The effects of diffusion and dispersion can be ignored.

(a) Determine the kinematic porosity of the aquifer.

[50%]

(b) Given that the K_d value for toluene in the same aquifer material is 0.086 cm^3 g^{-1}, when would you expect toluene to also be detected in the observation well.

[20%]

(c) The resulting concentration of toluene in the reservoir is 450 ppm. The toluene is expected to decay with a half-life of 126 days due to anaerobic microbial degradation. Assuming that water in the observation well is purely derived from the reservoir, determine the toluene concentration expected in the observation well.

[30%]

continued

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Formula sheet

Hydraulics

The continuity equation:

$$A_1 v_1 = A_2 v_2$$

The momentum equation:

$$F = \rho Q(v_2 - v_1)$$

Bernoulli equation:

$$E = \frac{p_1}{\rho g} + \frac{v_1^2}{2g} + z_1 = \frac{p_2}{\rho g} + \frac{v_2^2}{2g} + z_2$$

Reynolds' number:

$$\mathrm{Re} = \frac{4\rho v m}{\mu}$$

Hydraulic radius:

$$m = \frac{A}{P}$$

Darcy-Weisbach equation:

$$\frac{\partial E}{\partial s} = -\frac{f v^2}{2mg}$$

Chezy equation:

$$\frac{\partial E}{\partial s} = -\frac{v^2}{C^2 m}$$

Manning's equation:

$$\frac{\partial E}{\partial s} = -\frac{n^2 v^2}{m^{4/3}}$$

Partially filled circular pipe geometry:

$$A = \frac{D^2}{4}\left(\theta - \frac{\sin 2\theta}{2}\right)$$

$$P = \theta D$$

$$\theta = \arccos\left(1 - \frac{2H}{D}\right)$$

Contaminant hydrology

Darcy's law:

$$q \equiv \frac{Q}{A} = -K\frac{\partial h}{\partial s}$$

Hydraulic head:

$$h = \psi + z$$

Pressure head:

$$\psi = \frac{p}{\rho g}$$

Advective travel time:

$$t_a = \frac{L}{v}$$

continued

Pore-water velocity:

$$v = \frac{Q}{An_k}$$

Peclet number:

$$P_e = \left(\frac{D_E}{qL} + \frac{\alpha_L}{L} \right)^{-1}$$

Linear adsorption:

$$S = K_d C$$

Retardation factor:

$$R = 1 + \frac{\rho_b K_d}{n_k}$$

Exponential decay:

$$C = C_0 \exp(-\lambda t)$$

Half-life:

$$t_{1/2} = \frac{\ln 2}{\lambda}$$

END